Natalia A. Agapova

ML Engineer

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Education

Innopolis, Russia — **Innopolis University** — 2022 – 2026

- **Major:** BSc in Applied Artificial Intelligence (GPA: 4.02 / 5)

Employment

Croissan Studio — **Apr 2025 to present**

ML Software Engineer — *Apr 2026 to present*

- Fully fine-tuned transformer language models and adapted others through prompting for client products, comparing versions on task metrics.

ML Engineer & Product Designer — *Jul 2025 to Apr 2026*

- Trained and adapted generative image and talking-avatar models for client projects, judging output quality with metrics.
- Designed the product interfaces and client-facing flows around those models, covering both roles across the same sprints.

Software Engineer Intern — *Apr 2025 to Jul 2025*

- Worked with other engineers to ship these models inside production features, and explained results and limitations to clients and teammates.

- **Leveraged knowledge** in Python, PyTorch, Hugging Face Transformers, Diffusers, and LoRA fine-tuning.
- *Letter of recommendation from the studio's CEO* →

Asimov Lab (startup) — **Feb to Sep 2024**

Frontend Lead, ML Developer

- Led the frontend team at an AI test-generation startup while contributing to the ML side of a model that generates tests for teachers and students, from data and model evaluation to shipped UI, in close collaboration with the ML and product teams.

- **Leveraged knowledge** in React, TypeScript, Python, and scikit-learn.

Freelance — **2022 to present**

ML Engineer

- Applied ML work, including model development and evaluation, delivered alongside design and web development for independent clients.

- **Leveraged knowledge** in Python, PyTorch, and scikit-learn.

Software Projects

Fair Resume Screening — NLP · Explainability · Debiasing · Published, first author

- Fine-tuned BERT to sort resumes into 9 IT categories (HeadHunter data), measured by accuracy, by evaluating 39+ debiasing configurations with Integrated Gradients (Captum) bias audits.
- Found baseline accuracy of 60.9%, measured by counterfactual analysis, by swapping only the applicant's city and showing it flipped the model's prediction in 7.7% of pairs; this motivated the debiasing work. Published as first author.
- **Utilized:** Python, PyTorch, BERT, Captum, pandas.

Emotion Detection — Computer Vision · On-device Deployment

- Built a CNN image classifier on FER2013, measured at 58.7% accuracy / macro-F1 0.569 (up to 60.8% with ensembling and test-time augmentation), by quantizing the model to INT8 and deploying it to Core ML and ONNX Runtime Web at 1.15 ms latency.
- **Utilized:** Python, PyTorch, Core ML, ONNX Runtime.

Personal Knowledge System — RAG · NLP · Embeddings

- Built a retrieval-augmented assistant that answers strictly from a user's own PDFs, measured by filename+page citation accuracy, by chunking documents and embedding them with SentenceTransformer into ChromaDB, served through Ollama (Llama 3.2).
- **Utilized:** Python, SentenceTransformer, ChromaDB, Ollama.

Skills

ML / AI: *(proficient)*: Python, PyTorch, BERT, Captum, model evaluation & debiasing, RAG (ChromaDB, SentenceTransformer) *(familiar)*: LLMs (Ollama), OpenCV, MediaPipe, Core ML, ONNX Runtime
Tools: *(familiar)*: Figma, front-end development, Git/GitHub
Languages: Russian C2 (native) · English C1+ · Ukrainian B1 · French A2 · Tatar A2 · Korean A1

Publications

Agapova, N. A.; Lukmanov, R. A. "Improving Fairness in AI-Powered Recruitment: An Interpretable Resume Screening System", 2026. doi:10.66693/mathai.1017

Kholodov, Y. A.; Salloum, H.; Agapova, N. A. "Quantum Annealing Approaches to Breaking RSA Encryption", *Voprosy kiberbezopasnosti*, 2025. doi:10.21681/2311-3456-2025-4-127-133